

Backend Developer Interview Transcript

Sample transcript for InterviewAI testing

Interviewer: Hi, could you briefly introduce yourself?

Candidate: Sure. I'm a backend developer mainly working with Node.js, Express and MongoDB. I've built REST APIs, authentication systems and a few real-time applications using Socket.IO. I've also worked a little with Redis.

Interviewer: Great. Can you explain what Node.js is?

Candidate: Umm, Node.js is basically JavaScript that runs outside the browser. It uses the V8 engine and it's mainly used for creating backend applications and APIs.

Interviewer: Is Node.js single-threaded?

Candidate: Yes, Node.js is single-threaded, so it can only do one thing at a time. But it uses the event loop to handle multiple requests.

Interviewer: Can you explain the event loop in a little more detail?

Candidate: Yeah, so, umm, the event loop continuously checks whether there are asynchronous operations waiting to complete. When something finishes, its callback gets executed. I know there are different queues involved, but I'm not completely sure about how all of them work.

Interviewer: That's okay. What's the difference between authentication and authorization?

Candidate: Authentication means checking who the user is, like verifying their email and password. Authorization happens after authentication and determines what that user is allowed to do. For example, an admin may be able to delete users while a normal user cannot.

Interviewer: Good. Suppose your API becomes very slow because you're repeatedly fetching the same expensive data from the database. What might you do?

Candidate: I would probably use Redis. We could store frequently requested data in Redis so we don't have to query MongoDB every time. Umm, when the data changes we would also need to delete or update the cached value.

Interviewer: Tell me about a time you encountered a difficult bug in one of your projects.

Candidate: Yeah, I had a problem with Socket.IO where users sometimes weren't receiving events. I spent quite a lot of time debugging it. Eventually I found that some users weren't joining the correct document room. I fixed the room joining logic and then tested it with multiple browser windows.

Interviewer: What did you learn from that experience?

Candidate: I guess I learned that debugging real-time applications can be difficult and that I should check each part of the flow instead of randomly changing code.

Interviewer: Great. That's all from me. Do you have any questions?

Candidate: Yes. What would the first few months look like for a backend developer joining your team?